Instrumental Variables: IV, LATE and Weak Instruments

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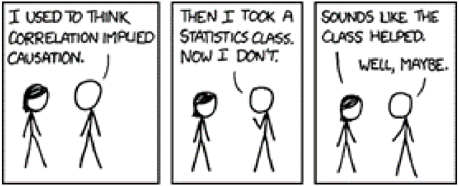

DAG intuition

- We observe a structural relationship:

$$y_i = \beta_0 + \beta_1 H_i + \varepsilon_i$$

- Problem: H_i is endogenous
- DAG intuition:
  - $Z \to H \to Y$
  - $H \leftrightarrow \varepsilon$
  - Valid IV requires:
    - Relevance: $Z \to H$
    - Exclusion: $Z \not\to Y$ except via H

Structural model

We start with:

$$y_i = a + bH_i + e_i \tag{1}$$

We are interested in b.

Conditional expectations by instrument status

Take expectations of (1) conditional on Z_i.

$$E[y_i \mid Z_i = 1] = a + bE[H_i \mid Z_i = 1] + E[e_i \mid Z_i = 1]$$

$$E[y_i \mid Z_i = 0] = a + bE[H_i \mid Z_i = 0] + E[e_i \mid Z_i = 0]$$

Difference in outcomes

Subtract:

$$E[y_i \mid Z_i = 1] - E[y_i \mid Z_i = 0]$$

$$= b\,(E[H_i \mid Z_i = 1] - E[H_i \mid Z_i = 0]) + (E[e_i \mid Z_i = 1] - E[e_i \mid Z_i = 0])$$

Key IV assumption

If:

$$E[e_i \mid Z_i = 1] = E[e_i \mid Z_i = 0]$$

then:

$$E[y_i \mid Z_i = 1] - E[y_i \mid Z_i = 0] = b\left(E[H_i \mid Z_i = 1] - E[H_i \mid Z_i = 0]\right)$$

Wald estimator

Solving for b:

$$\hat{b} = \frac{E[y_i \mid Z_i = 1] - E[y_i \mid Z_i = 0]}{E[H_i \mid Z_i = 1] - E[H_i \mid Z_i = 0]}$$

This is the **Wald estimator**.

Two-stage least squares

First stage:

$$H_i = \pi_0 + \pi_1 Z_i + v_i$$

Second stage:

$$y_i = \beta_0 + \beta_1 \hat{H}_i + u_i$$

Do the IV exercise now!

First task (get ready to speak about this):

- Find a paper that uses IV
- Answer the following:
  - What is the relation of interest and why is an instrument needed?
  - What is the evidence/argument for the IV requirements?
  - Bonus: do the results differ from OLS?

Second task:

- Replicate the main results of the famous Card (1993) distance to college instrument.
- Data and more instructions on github

IV works because the population is a mixture of groups

With a binary instrument $Z \in \{0, 1\}$ and treatment $D \in \{0, 1\}$, each individual has two potential treatment states:

$$D_1 \quad (\text{treatment if } Z = 1) \qquad D_0 \quad (\text{treatment if } Z = 0)$$

This gives four logically possible groups.

The four groups

- **Always takers:** $D_1 = 1, D_0 = 1$
  - Always treated, regardless of instrument
- **Never takers:** $D_1 = 0, D_0 = 0$
  - Never treated, regardless of instrument
- **Compliers:** $D_1 = 1, D_0 = 0$
  - Take treatment if and only if encouraged by Z
- **Defiers:** $D_1 = 0, D_0 = 1$
  - Do the opposite of the instrument (ruled out under monotonicity)

Monotonicity assumption (key restriction)

We assume:

$$D_1 \geq D_0$$

This rules out defiers.

- Interpretation: the instrument weakly increases treatment take-up
- Without this assumption, LATE is not identified in the standard way

Why this decomposition matters

The observed treatment rate is a mixture:

$$E[D \mid Z = 1] = P(\text{always takers}) + P(\text{compliers})$$

$$E[D \mid Z = 0] = P(\text{always takers})$$

So:

$$E[D \mid Z = 1] - E[D \mid Z = 0] = P(\text{compliers})$$

Key insight: IV only uses compliers

The Wald estimator:

$$\frac{E[Y \mid Z = 1] - E[Y \mid Z = 0]}{E[D \mid Z = 1] - E[D \mid Z = 0]}$$

can be rewritten as:

Average treatment effect for compliers

Intuition: who is actually driving the IV estimate?

- Always takers: unaffected by instrument $\rightarrow$ cancel out
- Never takers: unaffected by instrument $\rightarrow$ cancel out
- Defiers: ruled out
- Only compliers respond to Z

Key message

IV is a **selection device**: it isolates the subpopulation whose behavior changes when Z changes.

Why IV differs from OLS

OLS answers:

$$E[Y_1 - Y_0]$$

- if no endogeneity
- recall weighting result form L1+L2

IV answers:

$$E[Y_1 - Y_0 \mid \text{compliers}]$$

These are equal only under strong homogeneity assumptions

Economic meaning of compliers

Compliers are:

- Individuals on the margin of treatment adoption
- People whose decision is actually influenced by the instrument

Examples:

- Draft eligibility $\rightarrow$ people who serve only if drafted
- Twins $\rightarrow$ parents whose decision to have another child changes
- Quarter of birth $\rightarrow$ people affected by schooling laws

Important implication

IV is not a population parameter

It is a **mechanism-specific parameter**:

LATE depends on the instrument used

Different instruments $\rightarrow$ different compliers $\rightarrow$ different causal effect.

Complier analysis: intuition

- We cannot identify compliers at the individual level
- But we can characterize them at the group level
- This helps interpret the LATE:
  - Who is driving the IV estimate?
  - Why does IV differ from OLS?

Types of individuals

Under binary Z and D:

- Always takers: $D_1 = D_0 = 1$
- Never takers: $D_1 = D_0 = 0$
- Compliers: $D_1 > D_0$
- Defiers: ruled out (monotonicity)

Size of complier population

Key result:

$$P(\text{compliers}) = E[D \mid Z = 1] - E[D \mid Z = 0]$$

This is exactly the **Wald first stage**.

Compliers among treated

$$P(D_1 > D_0 \mid D = 1) = \frac{E[D \mid Z = 1] - E[D \mid Z = 0]}{P(D = 1)}$$

Interpretation:

- Share of treated who were actually moved by the instrument

Examples: size and share of compliers

Study	IV	First stage	Share treated
Angrist (1990)	Draft eligibility	0.534	0.101
Angrist & Evans (1998)	Twins	0.603	0.966
Angrist & Krueger (1991)	Quarter of birth	0.016	0.034

Characterising compliers

We can compare characteristics:

$$\frac{P(X = 1 \mid \text{complier})}{P(X = 1)} = \frac{\text{first stage for subgroup}}{\text{overall first stage}}$$

Interpretation:

- Who are compliers relative to population?

Twin instrument example

This is for Angrist & Evans (1998):

Variable	Population	Compliers	Ratio
Age: at least 30 first birth	0.0029	0.004	1.39
Black/Hispanic	0.125	0.103	0.822
College graduate	0.132	0.151	1.14

Stata/R packages (`ivdesc`) extend this to non-binary instruments.
The intuition is similar.

Key takeaway

- We do not observe compliers individually
- But we can describe them statistically
- This helps interpret LATE:
    - Not ATE
    - Not population-wide effect
- Note how this is similar to OLS (recall weeks 1+2) that requires conditional independence - again, we are just estimating effects for the subgroup that has remaining treatment variation.

Weak Instruments

- IV is consistent under standard assumptions, but can be biased in finite samples
- For a long time, this distinction was largely ignored in applied work
- Early 1990s literature highlighted a key issue:
  - instruments weakly correlated with endogenous regressors
  - many instruments used for one endogenous variable (overidentification problem)
- In the extreme case:
  - if there is no first stage at all
  - the 2SLS sampling distribution is centered on the probability limit of OLS

Setup

We consider a simple model with a single endogenous regressor:

$$y = \beta x + \nu$$

First stage:

$$x = Z'\pi + \eta$$

- Z = instrument(s)
- x is endogenous if ν and η are correlated

OLS bias under endogeneity

If we estimate by OLS:

$$E[\hat{\beta}_{OLS} - \beta] = \frac{\text{Cov}(\nu, x)}{\text{Var}(x)}$$

Using $x = Z'\pi + \eta$:

$$\text{Cov}(\nu, x) = \text{Cov}(\nu, \eta) = \sigma_{\nu\eta}$$

so:

$$E[\hat{\beta}_{OLS} - \beta] = \frac{\sigma_{\nu\eta}}{\sigma_x^2}$$

- OLS is biased whenever ν and η are correlated

Weak instruments and bias towards OLS

It can be shown that the bias of 2SLS is approximately:

$$E[\hat{\beta}_{2SLS} - \beta] \approx \frac{\sigma_{\nu\eta}}{\sigma_{\eta}^{2}} \cdot \frac{1}{F+1}$$

where:

- F is the population analogue of the first-stage F-statistic

Interpretation

- If $F \to 0$ (weak instruments):

$$E[\hat{\beta}_{2SLS} - \beta] \to \frac{\sigma_{\nu\eta}}{\sigma_\eta^2}$$

  - IV behaves like OLS

- If $F \to \infty$ (strong instruments):

$$E[\hat{\beta}_{2SLS} - \beta] \to 0$$

  - IV becomes approximately unbiased

Connection to OLS limit

Note:

- If there is no first stage ($\pi = 0$):

$$\sigma_x^2 = \sigma_\eta^2$$

- Then OLS bias:

$$\frac{\sigma_{\nu\eta}}{\sigma_x^2} = \frac{\sigma_{\nu\eta}}{\sigma_\eta^2}$$

- This equals the weak-IV limit of 2SLS bias

Key insight

When instruments are weak, 2SLS converges toward OLS in finite samples.

Why weak instruments are problematic

Recall IV is a ratio estimator:

$$\hat{\beta}_{IV} = \frac{\widehat{\text{reduced form}}}{\widehat{\text{first stage}}}$$

- If the first stage is small or noisy:
  - estimates become unstable
  - distribution becomes highly non-normal
  - large finite-sample bias arises

Weak Instruments: Adding More Instruments

- Adding more weak instruments can **increase the bias of 2SLS**
- Intuition: instruments with little or no predictive power dilute the first stage
- When we add instruments without meaningful explanatory power for x:
  - the first-stage fit does not improve
  - the effective signal-to-noise ratio declines
  - the first-stage F-statistic tends toward zero
- As a result:
  - weak-instrument bias increases
  - 2SLS estimates move closer to OLS

Just-identified vs overidentified IV

- **Just-identified IV**:
  - number of instruments = number of endogenous regressors
  - weak instrument bias is typically less severe
- **Overidentified IV**:
  - more instruments than endogenous variables
  - additional instruments may be weak
  - weak instruments accumulate and distort inference
- Important nuance (Angrist & Pischke):
  - IV is often described as “approximately unbiased”
  - this statement only holds when the first stage is sufficiently strong
  - otherwise finite-sample bias can be substantial

Evidence from the literature

- Bound, Jaeger and Baker (1995) show how weak instruments affect IV estimates
- They revisit the Angrist and Krueger (1991) study on returns to education
- Key idea: different valid instrument sets produce very different first-stage strength and estimates
- Example (Angrist & Krueger):
  - Classic study on returns to education using compulsory schooling variation
  - Instrument: quarter of birth (affects schooling due to compulsory schooling laws)
  - Key contribution: use of **many instruments via interactions**

Angrist & Krueger: intuition

- Many interaction-based instruments add little independent variation in x
- They increase the number of instruments mechanically but not their strength
- Consequences:
  - first-stage F-statistic declines
  - weak-instrument bias increases
  - 2SLS estimates move toward OLS

Many Instruments: Angrist & Krueger (1991)

- Instrument sets:
  - Baseline: quarter-of-birth dummies (3 instruments)
  - Expanded: interactions with state-of-birth and year-of-birth
  - Up to **180 instruments** in some specifications

	(1) OLS	(2) IV	(3) OLS	(4) IV	(5) OLS	(6) IV
Coefficient	.063 (.000)	.142 (.033)	.063 (.000)	.081 (.016)	.063 (.000)	.083 (.009)
F (excluded instruments)		13.486		4.747		2.428
Partial R^2 (excluded instruments, ×100)		.012		.043		.133
F (overidentification)		.932		.775		.919
		Age Control Variables				
Age, Age^2	x	x			x	x
9 Year of birth dummies			x	x		
		Excluded Instruments				
						x
Quarter of birth		x		x		x
Quarter of birth × year of birth				x		x
Number of excluded instruments		3		30		180

Key takeaway

- More instruments is not necessarily better
- Weak instruments can accumulate and weaken identification

Core message

Instrument proliferation can be as damaging as endogeneity itself.

- Practical implication:
  - prefer parsimonious sets of strong instruments
  - always report first-stage strength

Practical consequences

- IV estimates can be:
  - highly sensitive to specification
  - biased toward OLS
  - misleadingly significant or insignificant
- Standard asymptotic approximations may fail

Modern weak instrument diagnostics

- Classical rule-of-thumb: $F > 10$ (Staiger & Stock)
- More robust approach:
  - Montiel Olea & Pflueger (2013)
  - heteroskedasticity-robust first-stage test
  - allows clustering and autocorrelation
- Implication:
  - the relevant cutoff is closer to $F \approx 23$
  - weak IV problem is more severe than originally thought
- Practical tools:
  - `weakivtest` (Stata package)

The weak IV "arms race"

- Recent literature shows increasing stringency in what counts as "strong enough" instruments
- Lee et al. (2020):
  - focus not only on β but on the t-statistic
  - statistically significant IV estimates require much stronger first stages
  - sometimes $F \approx 100$ needed for reliable inference
- Key concern:
  - pre-testing on instrument strength distorts inference
  - similar to pre-testing in difference-in-differences trend tests

Is weak IV always a problem?

- Angrist & Kolesár (2022):
  - weak instruments are less of a concern in just-identified models
- Intuition:
  - bias exists, but inference distortion may be small
  - endogeneity would need to be extremely large for bias to dominate noise
- Key tension:
  - Lee et al. emphasize stronger first-stage requirements for inference
  - Angrist & Kolesár emphasize robustness in just-identified cases

Robust inference: Anderson–Rubin

- In just-identified or single-endogenous regressor settings:
- Anderson–Rubin confidence intervals:
  - remain valid regardless of instrument strength
  - do not rely on first-stage being strong
- Key advantage:
  - inference is robust even under weak instruments
- Practical implementation:
  - Chernozhukov & Hansen (2008)
  - available in Stata and R packages

Takeaways

- IV identifies causal effects using exogenous variation
- Wald estimator = reduced form / first stage
- 2SLS generalizes Wald
- IV identifies LATE (compliers)
- Weak instruments can invalidate inference - make sure to use appropriate packages

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